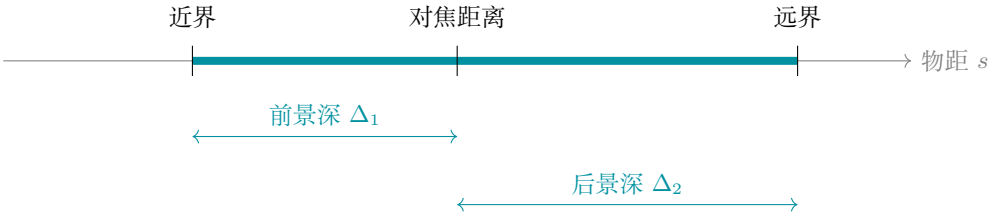

$$\Delta_1 = \frac{Nc\,s(s-f)}{f^2 + Nc(s-f)}, \qquad \Delta_2 = \frac{Nc\,s(s-f)}{f^2 - Nc(s-f)}$$

有限远界要求 $f^2 > Nc(s-f)$；否则远界为无穷

$$\mathrm{DOF} = \frac{2f^2 Nc\,s(s-f)}{f^4 - N^2c^2(s-f)^2}$$

$$s \gg f: \quad \mathrm{DOF} \approx \frac{2f^2 Nc\,s^2}{f^4 - N^2c^2s^2}$$

长度单位统一；N 为光圈数，c 为容许弥散圆，不预设为一像元

沿用薄透镜几何关系；示意不按距离比例绘制，不计算多片镜头与衍射